Thiruvenkadaprasath.S

192312016

EXPERIMENT NO:03

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Program:

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder

outlook = ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain',
        'Overcast', 'Sunny', 'Sunny', 'Rain', 'Sunny',
        'Overcast', 'Overcast', 'Rain']

temperature = ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool',
            'Cool', 'Mild', 'Cool', 'Mild', 'Mild',
            'Mild', 'Hot', 'Mild']

play = ['No', 'No', 'Yes', 'Yes', 'Yes', 'No',
     'Yes', 'No', 'Yes', 'Yes', 'Yes',
     'Yes', 'Yes', 'No']

le1 = LabelEncoder()
le2 = LabelEncoder()
le3 = LabelEncoder()

outlook_encoded = le1.fit_transform(outlook)
temperature_encoded = le2.fit_transform(temperature)
play_encoded = le3.fit_transform(play)

X = list(zip(outlook_encoded, temperature_encoded))
```

```
y = play_encoded

model = DecisionTreeClassifier(criterion='entropy')
model.fit(X, y)

new_sample = [[le1.transform(['Sunny'])[0], le2.transform(['Cool'])[0]]]

prediction = model.predict(new_sample)

print("New Sample: Outlook = Sunny, Temperature = Cool")

if prediction[0] == 1:
    print("Prediction: Play = Yes")
else:
    print("Prediction: Play = No")
```

output: